Independent Chiral Control in Theory-Space Models: A Rank-Preserving Framework and Its Application to Neutrino Mass Generation

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We develop a general framework of rank-preserving, element-wise matrix transformations for engineering fermion mass hierarchies in theory-space constructions. We prove that preservation of massless modes requires the transformation function to be separable, $g_f(i,j) = g_f^{(L)}(i)g_f^{(R)}(j)$, which in turn enables independent control of left- and right-chiral zero-mode profiles directly at the level of the theory-space mass matrix. This formalism unifies and extends the clockwork mechanism, permits controlled deformation of Kaluza–Klein spectra, and enhances hierarchy generation in GIM-like fine-cancellation scenarios. As a concrete application, we show that in theory-space models for neutrino masses, suitable transformations allow sub-eV light neutrinos to arise from TeV-scale new physics with only $\mathcal{O}(40)$ additional fermionic sites, while remaining consistent with charged-lepton flavor-violation bounds. In contrast, the corresponding untransformed models asymptote at the MeV scale and cannot access the phenomenologically required regime without extreme field multiplicities or hierarchical parameters.

I. INTRODUCTION

The origin of hierarchical scales in nature, from the electroweak scale to the sub-eV masses of neutrinos, remains a central and unresolved puzzle in particle physics. A particularly powerful class of Beyond the Standard Model (BSM) theories addresses this by *geometrizing* the hierarchy. In these "theory-space" models—including clockwork constructions and dimensional deconstruction—small effective couplings arise not from parameters inserted by hand, but dynamically from the localization of zero modes (massless fermionic states) across the sites of a discretized extra dimension [1–6]. The spatial profile of the zero mode directly determines the resulting hierarchical suppression. In addition to geometric setups, localization may also arise from randomness in the underlying parameters, leading to Anderson-like hierarchies [7–9].

Despite their success, standard theory-space constructions often possess a rigid structure. In many commonly studied models with nearest-neighbor interactions and simple boundary conditions, the left- and right-chiral zero-mode profiles are shaped by the same underlying Hamiltonian [2, 10–15]. As a result, the two chiralities are typically correlated, and reshaping them independently within a symmetric theory space Hamiltonian requires additional structure in conventional constructions. Independent chiral localization is a well-established feature of extra-dimensional models, here we emphasize that a similar independence can also be realized directly at the level of the theory space mass matrix through separable transformations. Moreover, in frameworks where light masses arise from fine-cancellation (GIM-like cancellation) between quasi-degenerate heavy states rather than from zero-mode localization, the accessible mass scales are often restricted by the asymptotic behaviour of the spectrum. These limitations motivate the search for a systematic method to reshape zero-mode profiles and modify spectral features while rigorously preserving the number of massless modes.

In this work we propose a general framework that achieves such control through a class of index-dependent, element-wise transformations of the form

$$b_{ij} = \frac{a_{ij}}{g_f(i,j)}\,. \tag{1}$$

These transformations may be viewed as a generalized Hadamard (Schur) scaling of the matrix entries [16], with broad relevance in areas such as network theory, image processing, and machine learning [17–19]. Our central result establishes a necessary and sufficient condition for these transformations to preserve the rank and nullity of the matrix, and hence the number of zero modes. We show that massless states are preserved *if and only if* the transformation function is separable,

$$g_f(i,j) = g_f^{(L)}(i)\,g_f^{(R)}(j)\,. \tag{2}$$

The separability condition plays a crucial role: the factors $g_f^{(L)}(i)$ and $g_f^{(R)}(j)$ act independently on rows and columns, allowing controlled re-sculpting of the left- and right-chiral zero-mode profiles while keeping the zero-mode count fixed.

This capability has significant consequences for theory-space model building. Specific choices of $g_f^{(L)}$ and $g_f^{(R)}$ reproduce familiar constructions such as exponential clockwork localization, warped-extra-dimensional–like profiles, or staggered patterns, while more general choices enable smooth interpolation between them. Crucially, the independent row/column action allows the chiral profiles of a Dirac fermion to be modified separately—a feature that is difficult to realize in conventional models where the zero-mode structure is tightly constrained by the underlying Hamiltonian. In addition, these transformations offer a systematic means of modulating the

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